

A linear–logarithmic positive-fraction bound from polynomial growth

Complete proof; independently reviewed

30 September 2026

Status. Full independent mathematical review by the root agent passed; this is not formal verification.

Let S be a finite set of generators of a group, without assuming $S = S^{-1}$. Let A_R consist of elements represented by positive words in S of length at most R , including the identity.

Theorem 1. *Suppose $|A_R| \leq (R + 1)^D$ for all integers $R \geq 0$, where $D \geq 1$. Every signed word of length $L \geq 1$ represents a fraction ab^{-1} , where a, b are positive words of lengths at most*

$$\lceil 16DL \log(8DL) \rceil.$$

The theorem gives existence; it does not assert an efficient algorithm for finding the two words.

Proof. Put $a_R = |A_R|$. For every $s \in S$, both sA_R and A_R are subsets of A_{R+1} and have cardinality a_R . Therefore

$$|sA_R \triangle A_R| \leq 2(a_{R+1} - a_R).$$

The same holds for s^{-1} by translation. If g has signed length L , the triangle inequality for symmetric difference, applied to its successive left translates, gives

$$|gA_R \triangle A_R| \leq 2L(a_{R+1} - a_R).$$

If $a_{R+1} - a_R < a_R/(2L)$, then gA_R intersects A_R (in fact more than half of each set lies in the intersection). Thus $gb = a$ for some $a, b \in A_R$, giving the required fraction.

It remains to find such an R . Set $T = \lceil 16DL \log(8DL) \rceil$. If none exists with $0 \leq R < T$, then

$$a_T \geq (1 + 1/(2L))^T \geq \exp(T/(4L)).$$

On the other hand $a_T \leq (T + 1)^D$. For $x = DL \geq 1$, the elementary bound $T + 1 \leq 18x \log(8x) \leq (8x)^2$ holds; the last inequality follows from $18 \log(8x) \leq 64x$, whose left-to-right ratio decreases for $x \geq 1$ and is already less than one at $x = 1$. Consequently

$$\log a_T \leq 2D \log(8DL) < T/(4L),$$

a contradiction. □

Corollary 1. *In the integral reduced free group on n generators, a signed word of length L has a positive fraction with both lengths*

$$O(D_n L \log(D_n L)), \quad D_n = \sum_{j=1}^n j \frac{n!}{(n-j)!} \leq en n!.$$

In particular, signed words of length $\exp(O(n \log n))$ can be cleared to positive fractions of the same asymptotic length.

Proof. The integral Magnus signature is injective. A positive word of length at most R has a nonnegative coefficient at each distinct-letter word of degree j , bounded by R^j . There are $n!/(n-j)!$ such coordinates. Therefore the number of possible signatures is at most

$$\prod_{j=1}^n (R^j + 1)^{n!/(n-j)!} \leq (R+1)^{D_n}.$$

Apply Theorem 1. The factorial estimate follows by changing variables to $n-j$ and bounding the resulting exponential series. $\square$

Effect on the dice problem. This improves the positive-fraction stage of the earlier quantitative saturation proof, which used the Malcev identity in a balanced product tree. Combined with the subsequently proved `order_doubling_prefix_completion.tex`, it yields an eventual fair-ray conductor $\exp(O(n \log n))$; see `factorial_order_eventual_conductor.tex`. This is an eventual-size result and does not provide an $\exp(O(n))$ minimal fair construction.